ภาคผนวก ค

Function in Libpcapnav which used in this project.

libpcapnav is a libpcap wrapper library that allows navigation to arbitrary locations in a tcpdump trace file between reads. The API is intentionally much like that of the pcap library. You can navigate in trace files both in time and space: you can jump to a packet which is at appr. 2/3 of the trace, or you can jump as closely as possible to a packet with a given timestamp, and then read packets from there. In addition, the API provides convenience functions for manipulating timeval structures.

Like libpcap, this library handles things through an opaque handle struct. For trace file navigation and reading packets, this handle is enough. If you need to apply BPF filters or write packets to disk, you can access the familiar pcap handle that is used internally

pcapnav_goto_offset ()

```
pcapnav_result_t pcapnav_goto_offset      (pcapnav_t *pn,
                                            off_t offset,
                                            pcapnav_cmp_t boundary);
```

The function tries to jump as closely as possible to a valid offset in the file near offset. The difference to pcapnav_set_offset() is that the latter simply modifies the file stream position, trusting that you know what you're doing. An offset of 0 is the first packet in the trace (i.e., the trace file header is not included). Using &boundary, you can limit the result to packets below, beyond, or anywhere around the requested offset.

pn : pcapnav handle.

offset : position to jump to.

boundary : where around the offset to jump to.

Returns : success state.

pcapnav_pcap ()

```
pcap_t*    pcapnav_pcap               (pcapnav_t *pn);
```

Use this function for interaction with libpcap. It returns the pcap handler for the given pcapnav handler. Do *not* mess with this unless you have to -- do not close the pcap handler etc behind pcapnav's back ...

pn : pcapnav handler.

Returns : pcap handler, or NULL on invalid input.